After the AP Computer Science Data Science Capstone

A Teacher Guide for a Data Investigation Project

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Preface

This resource is intended for high school computer science teachers seeking to extend student learning beyond AP Computer Science courses. It provides a teacher-facing guide for the After the AP Capstone.

The materials for the capstone are designed for students who have completed AP Computer Science Principles or AP Computer Science A, or who have comparable experience with text-based programming.

Introduction

Purpose

This resource is a teacher guide for a data investigation project designed to extend student learning beyond AP Computer Science courses. The guide supports high school computer science teachers in planning and facilitating a data science project that builds on students' prior programming experience and does not assume prior coursework in statistics or data science.

It explains the structure of the project, the dataset used, and the types of analyses students are expected to pursue. The project is intended for students who have completed AP Computer Science Principles or AP Computer Science A, or who have comparable experience with text-based programming.

> **Data Science Tools Used**
>
> This project uses Python with the `pandas` library for data manipulation (McKinney 2010) and the `matplotlib` library for data visualization (Hunter 2007).

Dataset

The project uses data from the General Social Survey (GSS). The GSS is primarily funded by the National Science Foundation (NSF) and conducted by the National Opinion Research Center (NORC), a non-partisan research organization. The GSS collects data on a variety of topics in contemporary American society to monitor and explain trends in opinions, attitudes, and behaviors.

The GSS contains a standard core of demographic, behavioral, and attitudinal questions, plus topics of special interest. Among the topics covered are education, civil liberties, national spending priorities, psychological well-being, and social mobility. It allows researchers to examine the structure and functioning of society in general, as well as the role played by relevant subgroups.

Dataset

General Social Survey

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About the General Social Survey

The GSS is a nationally representative series of cross-sectional interviews conducted in the United States since 1972. By adapting questions from earlier surveys, the GSS enables comparisons across decades. Originally proposed by James A. Davis, the survey has been administered by NORC at the University of Chicago and funded by the NSF since its inception. The GSS is currently designed by a set of Principal Investigators, with guidance from a Board of researchers.

In response to the COVID-19 pandemic, the 2021 GSS implemented methodological changes, including a shift to address-based sampling with a push-to-web, self-administered questionnaire. The 2022 GSS was designed as a bridge between the traditional face-to-face data collection used from 1972–2018 and the web-based approach introduced in 2021. It retained many questionnaire changes from the web mode while returning to a mixed-mode design that included face-to-face, web, and telephone data collection. The 2022 GSS was structured to facilitate comparison with the 2018 GSS and represents the first stage of a broader transition to a mixed-mode survey.

The Project Dataset

The dataset used for this project is a subset of the full GSS dataset, downloaded on December 31, 2025. The original dataset contained 46 variables and 75,699 observations.

Data Preparation & Cleaning

The dataset used in this project is based on the original GSS data but includes pre-processing steps to support classroom use and analysis.

The following changes were made prior to student use:

- A subset of variables and observations relevant to the project was selected.
- Non-substantive response codes (e.g., `.i: Inapplicable`, `.n: No answer`) were kept in their original form; some were removed or recoded during cleaning.
- The variable `age_group` was created.
- Variables originally labeled `cohort` and `year` were renamed to `birth_year` and `survey_year`.
- The variable `wwwhr` was recoded from categorical response values to numeric.
- The dataset was filtered to include respondents age 51 and under.

These steps were taken to reduce complexity while preserving the structure and integrity of the original survey data.

Teachers and students should be aware that the dataset is not a raw extract from the GSS, but a prepared version intended for instructional use.

Tidy and Tame Data Considerations

The project dataset has been prepared to follow principles of tidy data (Wickham 2014) and tame data (Kim, Ismay, and Chunn 2018) to support exploratory analysis.

In this dataset:

- Each row represents a single institution.
- Each column represents a single variable.
- Each cell contains a single, well-defined value.

Variables were also selected and organized to make the dataset *tame*, meaning, manageable, interpretable, and appropriate for classroom analysis. This includes limiting the number of variables, using consistent naming conventions, and providing variables that are meaningful for comparative analysis across institutions.

These design choices reduce unnecessary data wrangling and allow students to focus on asking questions, exploring patterns, and interpreting results, while still working with authentic, real-world data.

Variables

Names & Descriptions

This table provides brief descriptions of each variable used in the project. It is intended as a quick reference for understanding what each variable represents.

> **Variable Availability by Year**
>
> Unless otherwise noted, variables are available for survey years 1987, 1993, 2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2021, 2022, and 2024.

Variable Name	Description
`id`	Respondent ID number
`birth_year`	Respondent's year of birth
`survey_year`	Survey year for this respondent
`age`	Respondent's age
`age_group`	Respondent's age group
`sex`	Respondent's sex
`reg16`	Respondent's region of residence at age 16
`region`	Respondent's region at time of interview
`born`	Whether respondent was born in the United States
`parborn`	Whether respondent's parents were born in the United States
`granborn`	Number of respondent's grandparents born in the United States
`wrkstat`	Respondent's labor force status
`yearsjob`	Number of years respondent has been at current job
`moredays`	Number of days per month respondent works extra hours beyond usual schedule
`hrsrelax`	Number of hours respondent has to relax or pursue enjoyable activities

Variable Name	Description
`wwwhr`	Hours per week respondent spends browsing the internet (excluding email)
`tvhours`	Hours per day respondent spends watching television
`hsbio`	Whether respondent took a high school biology course
`hschem`	Whether respondent took a high school chemistry course
`hsmath`	Highest level of math completed by respondent in high school
	Whether respondent took a high school physics course
`hsclass1`	Respondent's favorite high school course
`dipged`	Type of high school credential earned by respondent
`educ`	Highest year of school completed by respondent
`degree`	Respondent's highest degree earned
`coldeg1`	Highest college degree earned by respondent
`majorcol`	Field of respondent's college degree
`colsci`	Whether respondent has taken any college-level science course
`whencol`	Year respondent received highest college degree
`voedcol`	Whether respondent completed non-college postsecondary education
`paeduc`	Highest year of school completed by respondent's father (or oldest same-sex parent)
`maeduc`	Highest year of school completed by respondent's mother (or youngest same-sex parent)
`padeg`	Highest degree earned by respondent's father (or oldest same-sex parent)
`madeg`	Highest degree earned by respondent's mother (or youngest same-sex parent)
`coneduc`	Respondent's confidence in the education system
`nateduc`	Opinion on improving the nation's education system

Variable Name	Description
`natenvir`	Opinion on improving and protecting the environment
`happy`	Respondent's general happiness
`happy7`	Respondent's self-reported happiness level
`hapunhap`	Whether respondent is happy or unhappy with life today
`goodlife`	Whether respondent believes their standard of living will improve
`opclout`	Belief that political connections are needed to get ahead
`ophrddwrk`	Belief that working hard is needed to get ahead
`opknow`	Belief that knowing the right people is needed to get ahead
`opwlth`	Belief that coming from a wealthy family is needed to get ahead
`opeduc`	Belief that a good education is needed to get ahead

Categories & Years

This table groups variables by topic area and indicates the survey years in which each variable is available. It can be used to identify which variables support comparisons across time.

Category	Variable	Years Available
Identifiers & Administrative		
	`id`	All
	`survey_year`	All
	`birth_year`	All
Age & Demographics		
	`age`	All
	`age_group`	All
	`sex`	All
Geographic Background		
	`region`	All
	`reg16`	All
Nativity & Family Background		

Category	Variable	Years Available
	born	All
	parborn	All
	granborn	All
Employment, Work, & Time Use		
	wrkstat	All
	yearsjob	2002, 2006, 2010, 2014, 2018, 2022
	moredays	2002, 2006, 2010, 2014, 2018, 2022
	hrsrelax	2002, 2006, 2010, 2014, 2018, 2022
Technology & Media Use		
	wwwhr	2000, 2002, 2004, 2006, 2010, 2012, 2014, 2016, 2018, 2021, 2022
	tvhours	All
Education (Respondent)		
	educ	2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2021, 2022, 2024
	degree	2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2021, 2022, 2024
	coldeg1	2006, 2008, 2010, 2012, 2014, 2016, 2018
	majorcol	2006, 2008, 2010, 2012, 2014, 2016, 2018
	colsci	2006, 2008, 2010, 2012, 2014, 2016, 2018
	whencol	2012, 2014, 2016
	voedcol	2012, 2014, 2016
	dipged	2012, 2014, 2016, 2018, 2021, 2022, 2024
	hsclass1	1993
Education (High School Coursework)		
	hsbio	2006, 2008, 2010, 2012, 2014, 2016, 2018
	hschem	2006, 2008, 2010, 2012, 2014, 2016, 2018

Category	Variable	Years Available
	`hsmath`	2006, 2008, 2010, 2012, 2014, 2016, 2018
	`hsphys`	2006, 2008, 2010, 2012, 2014, 2016, 2018
Education (Parental Background)		
	`paeduc`	All
	`maeduc`	All
	`padeg`	All
	`madeg`	All
Education & Public Institutions (Attitudes)		
	`coneduc`	All
	`nateduc`	All
Environment & Public Policy Attitudes		
	`natenvir`	All
Happiness & Well-Being		
	`happy`	All
	`happy7`	2002, 2012, 2022
	`hapunhap`	2002, 2008, 2018
	`goodlife`	All
Social Mobility & Opportunity Beliefs		
	`opclout`	1987, 2021
	`ophrddwrk`	1987, 2021
	`opknow`	1987, 2000, 2021
	`opwlth`	1987, 2000, 2021
	`opeduc`	1987, 2021

Survey Questions & Years

This table maps each variable to the original survey question and lists the years in which the question was asked. It helps clarify how variables were measured and when they appear in the dataset.

Variable	Survey Question	Years Asked
`id`	Administrative identifier; not a survey question.	All
`birth_year`	Derived from age and survey year; not asked directly.	All
`survey_year`	Administrative variable indicating year of interview.	All
`age`	What is your age?	All
`age_group`	Derived from respondent age; not asked directly.	All
`sex`	What is your sex?	All
`reg16`	In what region did you live when you were 16 years old?	All
`region`	In what region do you currently live?	All
`born`	Were you born in the United States?	All
`parborn`	Were your parents born in the United States?	All
`granborn`	How many of your grandparents were born outside the United States?	All
`wrkstat`	What is your current labor force status?	All
`yearsjob`	How many years have you been working at your current job?	2002, 2006, 2010, 2014, 2018, 2022
`moredays`	About how many days per month do you work extra hours beyond your usual schedule?	2002, 2006, 2010, 2014, 2018, 2022
`hrsrelax`	About how many hours do you have to relax or pursue activities that you enjoy?	2002, 2006, 2010, 2014, 2018, 2022
`wwwhr`	On average, how many hours per week do you use the Internet for browsing, not counting email?	2000, 2002, 2004, 2006, 2010, 2012, 2014, 2016, 2018, 2021, 2022
`tvhours`	On average, how many hours per day do you watch television?	All

Variable	Survey Question	Years Asked
`hsbio`	Did you take a biology course in high school?	2006, 2008, 2010, 2012, 2014, 2016, 2018
`hschem`	Did you take a chemistry course in high school?	2006, 2008, 2010, 2012, 2014, 2016, 2018
`hsmath`	What is the highest level of mathematics you completed in high school?	2006, 2008, 2010, 2012, 2014, 2016, 2018
`hsphys`	Did you take a physics course in high school?	2006, 2008, 2010, 2012, 2014, 2016, 2018
`hsclass1`	When you were in high school, what was your favorite course or subject?	1993
`dipged`	Which best describes the credential you received from high school?	2006, 2008, 2010, 2012, 2014, 2016, 2018
`educ`	What is the highest year of school you have completed?	2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2021, 2022, 2024
`degree`	What is the highest degree you have earned?	2000, 2002, 2004, 2006, 2008, 2010, 2012, 2014, 2016, 2018, 2021, 2022, 2024
`coldeg1`	What is the highest college degree you have earned?	2006, 2008, 2010, 2012, 2014, 2016, 2018
`majorcol`	What was your major or field of study for your college degree?	2006, 2008, 2010, 2012, 2014, 2016, 2018
`colsci`	Have you ever taken a college-level science course?	2006, 2008, 2010, 2012, 2014, 2016, 2018
`whencol`	In what year did you receive your college degree?	2012, 2014, 2016
`voedcol`	Have you completed any non-college postsecondary education?	2012, 2014, 2016
`paeduc`	What is the highest year of school completed by your father (or oldest same-sex parent)?	All
`maeduc`	What is the highest year of school completed by your mother (or youngest same-sex parent)?	All

Variable	Survey Question	Years Asked
`padeg`	What is the highest degree earned by your father (or oldest same-sex parent)?	All
`madeg`	What is the highest degree earned by your mother (or youngest same-sex parent)?	All
`coneduc`	How much confidence do you have in the education system?	All
`nateduc`	Are we spending too much, too little, or about the right amount on improving the nation's education system?	All
`natenvir`	Are we spending too much, too little, or about the right amount on improving and protecting the environment?	All
`happy`	Taken all together, how would you say things are these days?	All
`happy7`	Taking all things together, how happy would you say you are these days?	2002, 2012, 2022
`hapunhap`	Would you say you are generally happy or unhappy with your life today?	2002, 2008, 2018
`goodlife`	Do you think your standard of living will get better, worse, or stay about the same?	All
`opclout`	How important is having political connections for getting ahead in life?	1987, 2021
`ophrddwrk`	How important is working hard for getting ahead in life?	1987, 2021
`opknow`	How important is knowing the right people for getting ahead in life?	1987, 2000, 2021

Variable	Survey Question	Years Asked
`opwlth`	How important is coming from a wealthy family for getting ahead in life?	1987, 2000, 2021
`opeduc`	How important is having a good education for getting ahead in life?	1987, 2021

Missing & Non-Substantive Codes

This section describes codes used to represent missing, inapplicable, or non-substantive responses in the GSS. Missing and non-substantive codes appear in survey data when a question does not apply to a respondent, was skipped, or could not be answered. These values are not errors.

Some missing values occur because a question was never asked (for example, it was not included in a particular survey year), while others reflect meaningful responses such as "don't know" or "not applicable."

When analyzing the data, these codes should be handled intentionally. In most cases, they should be excluded from summaries and visualizations, rather than treated as valid responses.

Code	Meaning	Notes & When It Appears
`NaN`	Missing value (numeric)	Pandas missing value; appears in numeric columns
`.i: Inapplicable`	Inapplicable	Question does not apply due to skip patterns or survey design
`.n: No answer`	No answer	Question was asked, but no valid response recorded
`.d: Do not Know/Cannot Choose`	Don't know / cannot choose	Respondent could not or would not provide a substantive answer
`.s: Skipped on Web`	Skipped on web	Respondent skipped the question in web-based surveys
`.y: Not available in this year`	Not asked in this year	Variable not included in that survey year

Working with Missing Values

Missing and non-substantive codes are not errors. They indicate when a question did not apply, was skipped, or could not be answered. These values should usually be excluded from summaries and visualizations.

Helpful Notebook Tips

Introduction to Jupyter Notebook

A Jupyter Notebook is a web-based interactive computing environment that supports programming in Python, R, and other languages (Stokes and Harding 2025). In this project, you will use Jupyter Notebooks to write code, explore data, and document your analysis.

Overview of a Jupyter Notebook

Jupyter notebooks allow users to create and share documents that combine code, equations, visualizations, and formatted text (Stokes and Harding 2025). They are commonly used for data science because users can write and run code step by step, document their thinking, and view results directly alongside the code that produced them.

> **Avoid Multiple Tabs**
>
> Avoid opening multiple notebooks in separate tabs, as this can lead to memory overload, forcing you to restart your session.

Code & Text Cells

A Jupyter Notebook has three types of cells. For this project, you will use only two:

- **Code** cells for writing and running Python code
- **Markdown** or text cells for entering written responses.

Running Cells in a Jupyter Notebook

In a Jupyter Notebook, you run cells one at a time to see code output or display formatted text. To run a cell, click inside the cell and then click the **Run** button in the toolbar at the top of the notebook.

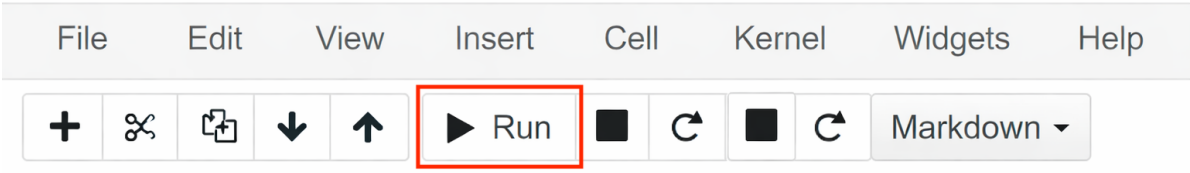

Figure 1: **Run** button in the Jupyter toolbar

You can also use keyboard shortcuts:

- PCs and Chromebooks: Hold **Ctrl** and press **Enter**
- Macs: Hold **Command** ( ) and press **Enter**

Adding Cells

Use the + button in the top toolbar to add a new cell.

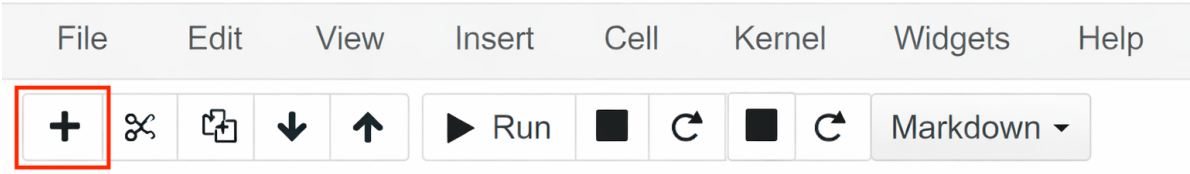

Figure 2: + button in the Jupyter toolbar

Then set the cell type as needed:

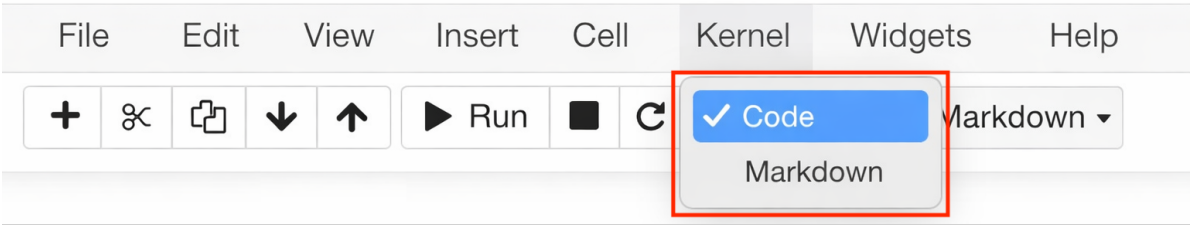

Figure 3: **Code** or **Markdown** button in the Jupyter toolbar

- Choose **Code** for code
- Choose **Markdown** for text

Running All Cells from Top Down

The **Save** button saves your code and text, but it *does not* save data stored in memory. Whenever you reopen a notebook, you should re-run all code cells from the top.

To do this, go to **Kernel** → **Restart & Run All**.

> **Restart & Run All**
>
> This option restarts the notebook and runs every cell in order from top to bottom. If a code cell encounters an error, execution will stop and later cells will not run.

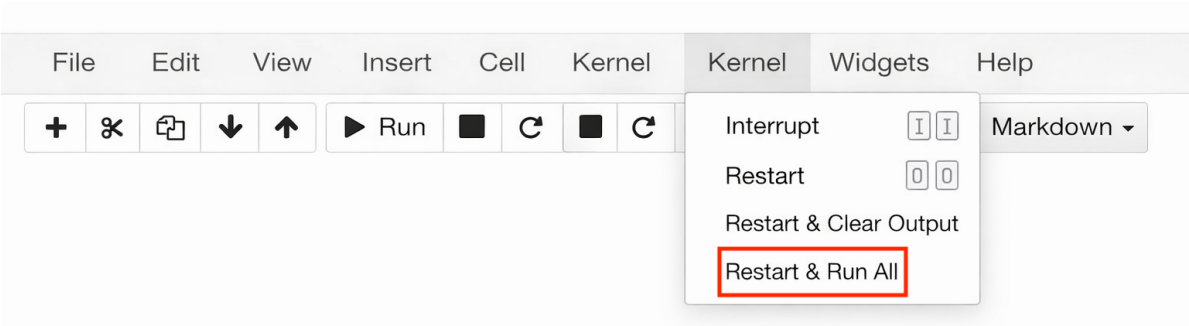

Figure 4: **Restart & Run All** button in the Jupyter toolbar

Useful Python Commands

```
import sys
print("PYTHON:", sys.executable)

import pandas as pd
print("PANDAS:", pd.__version__)
```

```
PYTHON: /Users/mharding/anaconda3/envs/datascience/bin/python
PANDAS: 1.5.3
```

Notebook 1: Basic Python & Data Structures

Variables and Assignment

Python variables are created by assignment. A valid Python variable name must start with a letter or an underscore, use only letters, numbers, and underscores, and can't be a reserved Python keyword.

Syntax

```
variable_name = value
```

Example

```
x = 10
x
```

```
10
```

`print()`

The `print()`function displays output.

Syntax

```
print(value)
```

Example

```
print("Hello!")
print(x)
```

```
Hello!
10
```

Lists (`[ ]`)

Lists store ordered collections of values. Items are accessed by index, starting at 0.

Syntax

```
my_list = [item1, item2, item3]
my_list[index]
```

Example

```
letters = ["A", "B", "C", "D", "E"]

print(letters[0])
print(letters[1])
print(letters[2])
print(letters[3])
print(letters[4])
```

```
A
B
C
D
E
```

List Slices (:)

Slices select a range of values from a list. The ending index is not included.

Syntax

```
my_list[start:end]
my_list[:end]
my_list[start:]
```

Example (First 3 values)

```
print(letters[0:3])
print(letters[:3])
print(letters[2:])
```

```
['A', 'B', 'C']
['A', 'B', 'C']
['C', 'D', 'E']
```

Example (Last 3 values)

```
print(letters[2:])
print(letters[-3:])
```

```
['C', 'D', 'E']
['C', 'D', 'E']
```

> Slice reminder
>
> - `[:3]` means items at index positions 0–2
> - `[-3:]` means the last three items

`for`

A `for` loop repeats an action for each item in a collection.

Syntax

```
for item in collection:
    print(item)
```

Example

```
for item in letters[-3:]:
    print(item)
```

```
C
D
E
```

`with open()`

This pattern reads a file and automatically closes it afterward.

Syntax

```
my_list = []

with open("path/to/file.txt", "r") as file:
    for line in file:
        clean = line.strip()
        my_list.append(clean)
```

Example

```
education = []

with open("data/education.txt", "r") as file:
    for line in file:
        newline = line.strip()
        education.append(newline)

education[:10]
```

```
['11th grade',
 '4 years of college',
 '4 years of college',
```

```
 '3 years of college',
 '5 years of college',
 '11th grade',
 '12th grade',
 '12th grade',
 '6 years of college',
 '6 years of college']
```

> **Why `.strip()` matters**
>
> Text files often contain invisible characters such as `"\n"` at the end of each line. `.strip()` removes extra whitespace from values.

`.append()`

The `.append()` method adds one item to the end of a list.

Syntax

```
my_list.append(value)
```

Example

```
print(letters)
letters.append("F")
print(letters)
```

```
['A', 'B', 'C', 'D', 'E']
['A', 'B', 'C', 'D', 'E', 'F']
```

`len()`

The `len()` function returns the number of items in a collection.

Syntax

```
len(collection)
```

Example

```
print(letters)
print(len(letters))
```

```
['A', 'B', 'C', 'D', 'E', 'F']
6
```

set()

A set stores only unique values.

Syntax

```
set(collection)
```

Example

```
set(education)
```

```
{'.d:  Do not Know/Cannot Choose',
 '1 year of college',
 '10th grade',
 '11th grade',
 '12th grade',
 '2 years of college',
 '2nd grade',
 '3 years of college',
 '4 years of college',
 '4th grade',
 '5 years of college',
 '5th grade',
 '6 years of college',
 '6th grade',
 '7 years of college',
 '7th grade',
 '8 or more years of college',
 '8th grade',
 '9th grade',
 'No formal schooling'}
```

> **When to use `set()`**
>
> Use `set()` to identify distinct values in a list.

Dictionaries (`{ }`)

Dictionaries store key–value pairs. Values are accessed using keys.

Syntax

```
my_dict = {key1: value1, key2: value2}
my_dict[key1]
my_dict[key2]
```

Example (string keys)

```
print(exchange_rates)
print(exchange_rates["EUR"])
print(exchange_rates["JPY"])
print(exchange_rates["GBP"])
```

```
{'EUR': 0.92, 'JPY': 148.5, 'GBP': 0.79}
0.92
148.5
0.79
```

Example (numeric keys)

```
print(phase_changes)
print(phase_changes[-1])
print(phase_changes[0])
print(phase_changes[100])
```

```
{-1: 'Below freezing', 0: 'Freezing point of water', 100: 'Boiling point of water'}
Below freezing
Freezing point of water
Boiling point of water
```

> **Lists vs Dictionaries**
>
> - Use a list when order matters (index-based).
> - Use a dictionary when you need to look up values by a meaningful label (key-based).

Notebook 2: Basic Python & Data Structures

`import pandas as pd`

Pandas is a Python library used for working with tabular data.

Syntax

```
import pandas as pd
```

- `as pd` creates a shortcut (alias)
- `pd` is the standard alias for `pandas`

`pd.read_csv()`

Reads a CSV file and stores it as a `DataFrame`.

Syntax

```
dataframe_name = pd.read_csv("file_path.csv")
```

Example

```
gss = pd.read_csv("data/gss.csv")
```

- Creates a `pandas DataFrame` named `gss`.

Notebook 2 Pandas and Basic Dat…

Notebook 3: Visualizing Distributions

Notebook 4: Visualizing Trends

Notebook 5: Exploring Relationships and Trends

Resources

Video Resources

The General Social Survey (GSS) YouTube channel has an educational video series designed to support instructors, students, and researchers using GSS data.

The following videos provide an overview of the General Social Survey (GSS), from survey design and sampling to data access and instructional use.

Video	Description
Introduction to the GSS	Brief overview of the GSS, including its history, key stakeholders, and selected academic contributions. Positions the GSS as a widely used data resource for undergraduate instruction and research.
GSS Project Process & Life Cycle	Explains the full lifecycle of the GSS survey, including proposal review, testing, programming, pilot studies, data collection, cleaning, dissemination, and user support.
The GSS Structure & Basics	Describes the structure of the GSS, including core content, experiments, cross-sectional and panel studies, ISSP modules, and the balloting process. Covers basic skills for navigating, downloading, and interpreting GSS data.
Sampling in the GSS	Explains how representative samples are drawn for the GSS and introduces key sampling concepts used in the survey.
Total Survey Error in the GSS	Discusses sources of error in survey data, including nonresponse error, coverage error, sampling error, and measurement error.

Video	Description
GSS Wording Experiments	Provides examples of how question wording affects responses, highlighting GSS contributions to understanding measurement effects (e.g., national spending priority items).
The GSS Data Explorer	Tutorial on using the GSS Data Explorer, including online analytics, visualizations, and features introduced in Data Explorer 2.0.
GSS for Educators	Targeted to instructors; shares ideas for incorporating GSS data into lesson plans and classroom activities.
GSS for STEM	Focuses on working with data beyond survey responses, including paradata and linked datasets such as the death index.
Taking a Deep Dive into GSS Content	Overview of major variable groups and content areas, highlighting resources and recent trends relevant to researchers and educators.

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Partners

CourseKata

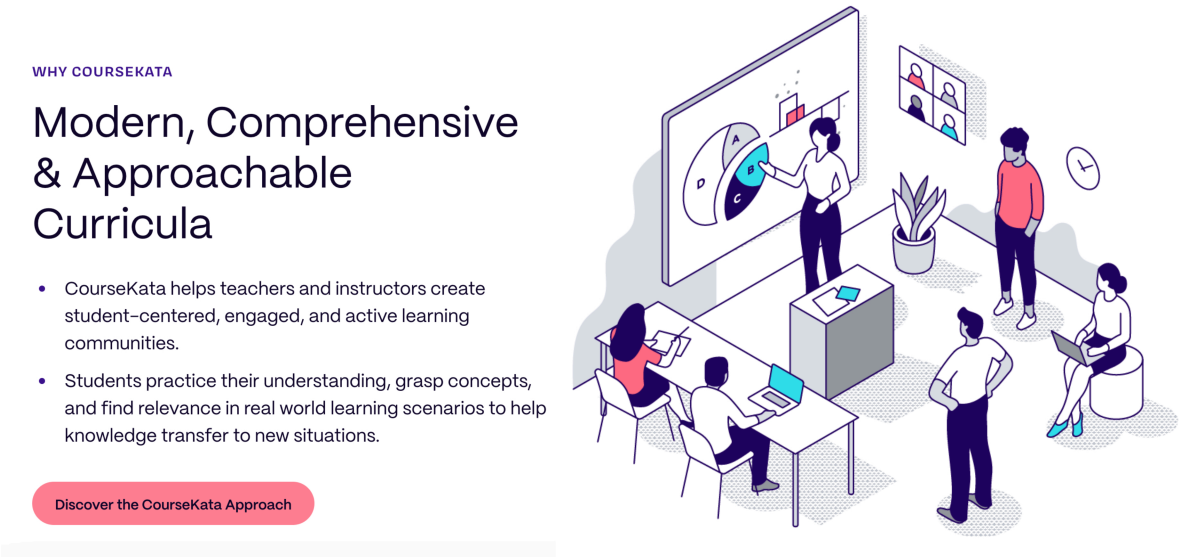

Figure 1: CourseKata

The After the AP Data Science Challenge uses Jupyter notebooks hosted on CKHub, developed by CourseKata. If this challenge sparked your interest, contact us CourseKata.org for more classroom-ready resources like these, including materials you can use throughout the year to supplement your regular curriculum.

We're a nonprofit team of educators and learning scientists who build both full-year and supplementary statistics and data science curricula using the same kinds of Jupyter notebooks, real datasets, and modeling tools used in the challenge.

Our technology and curriculum are designed to bring authentic data science practices into classrooms in ways that are teaching-friendly, meaningful, and equitable. Our research on lowering barriers to participation for students historically marginalized in STEM directly informs how we design our learning materials so more students can engage in mathematics that matters in the modern world.

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